

CST 304 -Computer Graphics & Image Processing

Module - 4

Fundamentals of Digital Image Processing

Text:Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing. Pearson, 4e, 2017

SYLLABUS

- Introduction to Image Processing and applications.
- Image as 2D data. Image representation in Gray scale, Binary and Colour images.
- Fundamental steps in Image Processing. Components of image processing system. Coordinate conventions. Sampling and quantization. Spatial and Gray Level Resolution.
- Basic relationship between pixels- Neighbourhood, Adjacency, Connectivity.
- Fundamentals of spatial domain- Convolution operation.

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IMAGE PROCESSING – Fundamentals

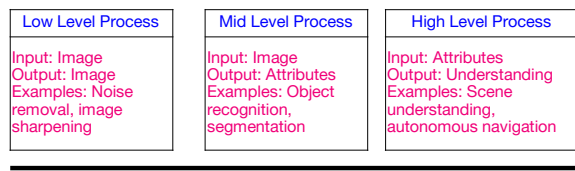
- Image processing involves processing or altering an existing image in a desired manner.
- Digital Image Processing - Two major tasks :
 - Improvement of pictorial information for human interpretation
 - Processing of image data for storage, transmission and representation for autonomous machine perception
- Need(Purpose) of image processing:-
 - Since the digital image is “invisible” it must be prepared for viewing on one or more output device (laser printer, monitor, etc)
 - The digital image can be optimized for the application by enhancing or altering the appearance of structures within it (based on: body part, diagnostic task, viewing preferences, etc)
 - It might be possible to analyze the image in the computer and provide cues to the radiologists to help detect important/suspicious structures (e.g.:Computed Aided Diagnosis, CAD)

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Image Processing Fields

- ❖ Computer Graphics: Creation of images
- ❖ Image Processing: Enhancement or other manipulation of the image
- ❖ Computer Vision: Analysis of the image content

The continuum from image processing to computer vision can be broken up into low, mid and high-level processes

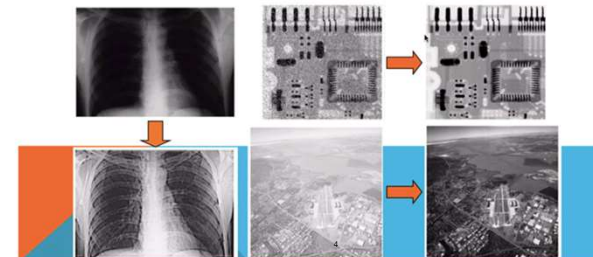


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EXAMPLE

Image Enhancement

One of the most common uses of DIP techniques: improve quality, remove noise etc



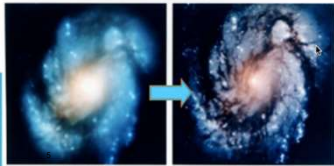
EXAMPLE

The Hubble Telescope

Launched in 1990 the Hubble telescope can take images of very distant objects

However, an incorrect mirror made many of Hubble's images useless

Image processing techniques were used to fix this



EXAMPLE

Artistic Effect

Artistic effects are used to make images more visually appealing, to add special effects and to make composite images

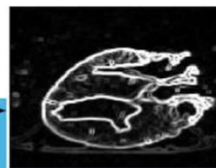
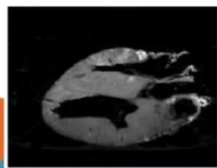


EXAMPLE

Medicine

Take slice from MRI scan of canine heart, and find boundaries between types of tissue

- Image with gray levels representing tissue density
- Use a suitable filter to highlight edges



Original MRI Image of a Dog Heart

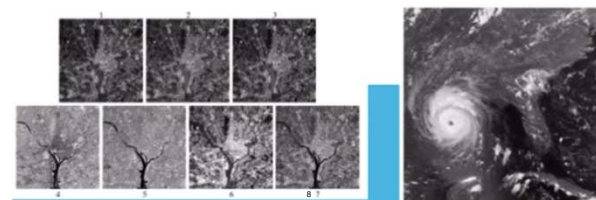
Edge Detection Image

EXAMPLE

GIS

Geographic Information Systems

- Digital image processing techniques are used extensively to manipulate satellite imagery
- Terrain classification
- Meteorology



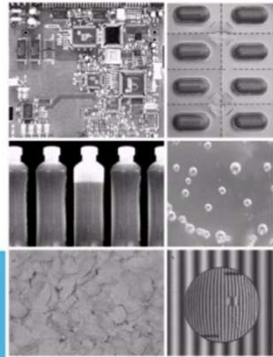
EXAMPLE

Industrial Inspection

Human operators are expensive, slow and unreliable

Make machines do the job instead

Industrial vision systems are used in all kinds of industries

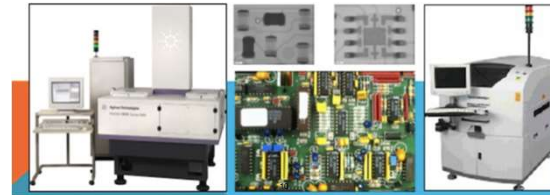


EXAMPLE

PCB Inspection

Printed Circuit Board (PCB) Inspection

- Machine inspection is used to determine that all components are present and that all solder joints are acceptable
- Both conventional imaging and x-ray imaging are used



EXAMPLE

Law Enforcement

Image processing techniques are used extensively by law enforcers

- Number plate recognition for speed cameras/automated toll systems
- Fingerprint recognition
- Enhancement of CCTV images



EXAMPLE

HCI

Try to make human computer interfaces more natural

- Face recognition
- Gesture recognition



Digital Image Processing Applications

- 1) Gamma Ray Imaging
 - Astronomical observations
 - Medical applications
- 2) X-ray Imaging
 - Medical diagnostics
 - Angiography
 - Computerized axial tomography (CT scan)
 - Industry applications
- 3) Ultraviolet Imaging
 - Lithography
 - Industrial inspection
 - microscopy
 - lasers
 - astronomical observations.

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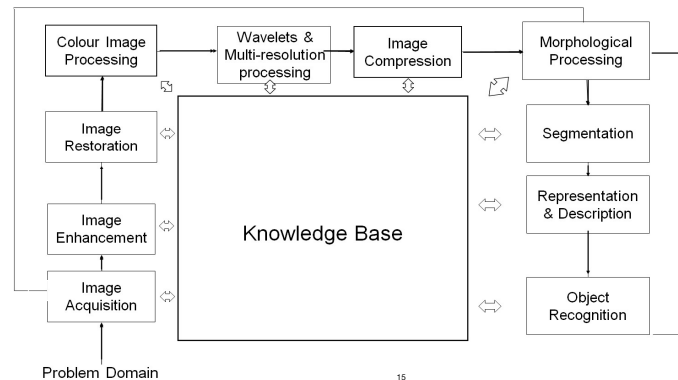
Digital Image Processing Applications

- 4) Imaging in the visible and Infrared band
 - light microscopy
 - astronomy
 - remote sensing
 - industry
 - law enforcement (biometrics)
- 5) Imaging in the Microwave Band
 - Radar —collect data over virtually any region at any time , regardless of weather or ambient lighting conditions
- 6) Imaging in the Radio Band
 - Medicine —magnetic resonance imaging (MRI)
 - Astronomy

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Fundamental Steps in Digital Image Processing

Outputs of these processes generally are images



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Outputs of these processes generally are image attributes

Fundamental Steps in Digital Image Processing

Step 1 : Image Acquisition

- The image is captured by a **sensor** (eg. Camera), and **digitized** if the output of the camera or sensor is not already in digital form, using analog-to-digital convertor.

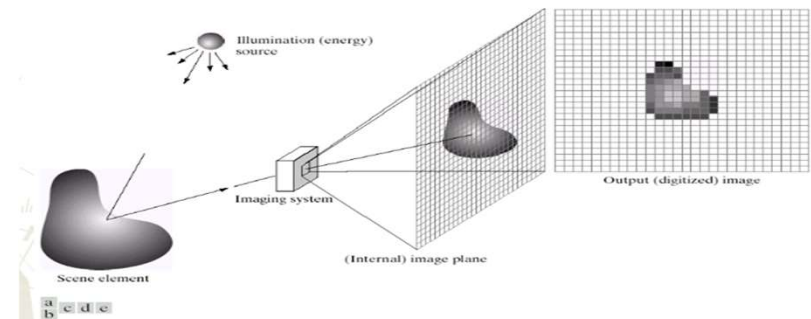
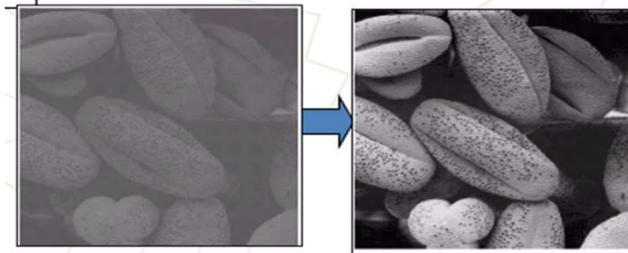


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Fundamental Steps in Digital Image Processing

Step 2 : Image Enhancement

- The process of manipulating an image so that the result is more suitable than the original for specific applications.
- The idea behind enhancement techniques is to bring out details that are hidden, or simply to highlight certain features of interest in an image.

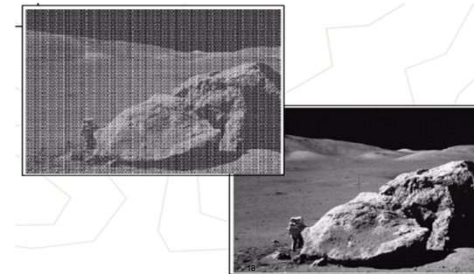


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Fundamental Steps in Digital Image Processing

Step 3 : Image Restoration

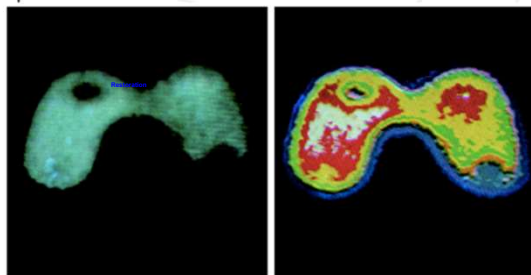
- Restoration** - Improving the appearance of an image. Tend to be mathematical or probabilistic models. (objective procedure)
- Enhancement**, on the other hand, is based on human subjective preferences regarding what constitutes a “good” enhancement result.



Fundamental Steps in Digital Image Processing

Step 4 : Colour Image Processing

- Use the colour of the image to extract features of interest in an image



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Fundamental Steps in Digital Image Processing

Step 5 : Wavelets(both frequency and time info are available)

- Are the foundation of representing images in various degrees of resolution.
- It is used for image data compression.

Step 6 : Compression

- Techniques for reducing the storage required to save an image or the bandwidth required to transmit it.

Name	Type	Modified	Size	Ratio	Ratio %
imprcomp1.zip	Image	21/09/2005 09:22	12,542	1%	12.4%
error.png	Image	07/04/2005 10:30	202	12%	2%
impr.png	Image	07/04/2005 10:30	202	12%	2%
impr2.png	Image	01/04/2005 10:10	5,153	0%	5.1%
impr3.png	Image	07/04/2005 10:30	206	14%	2%
impr4.png	Image	07/04/2005 10:30	205	13%	2%
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impr100.png	Image	07/04/2005 10:30	206	13%	2%

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Fundamental Steps in Digital Image Processing

Step 7 : Morphological Processing

- Tools for extracting image components that are useful in the representation and description of shape.
- In this step, there would be a transition from processes that output images, to processes that output image attributes.

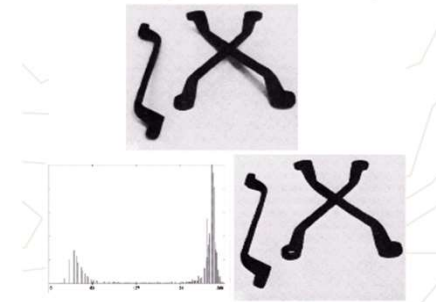


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Fundamental Steps in Digital Image Processing

Step 8 : Image Segmentation

- Segmentation procedures partition an image into its constituent parts or objects.

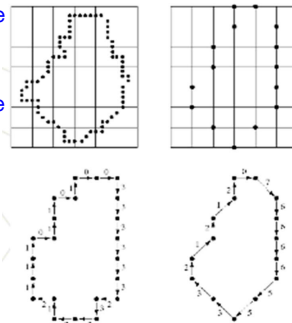


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Fundamental Steps in Digital Image Processing

Step 9 : Representation and Description

- Representation:** Make a decision whether the data should be represented as a boundary or as a complete region. It almost always follows the output of a segmentation stage.
- Boundary Representation:** Focus on external shape characteristics, such as corners
- Region Representation:** Focus on internal properties, such as texture
- Choosing a representation is only part of the solution for transforming raw data into a form suitable for subsequent computer processing (mainly recognition).
- Description:** also called, feature selection, deals with extracting attributes that result in some information of interest.

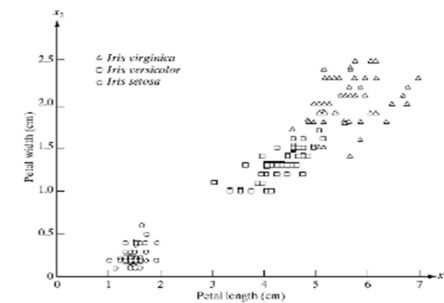


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Fundamental Steps in Digital Image Processing

Step 10: Object Recognition

- Recognition:** the process that assigns label to an object based on the information provided by its description.

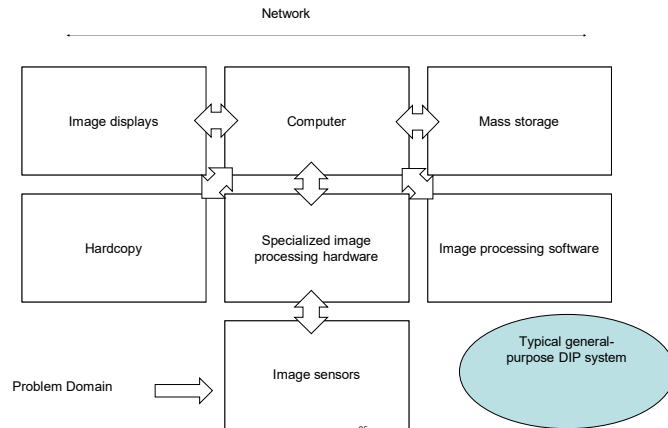


Step 11 : Knowledge Base

- Knowledge about a problem domain is coded into an image processing system in the form of a knowledge database.

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Components of an Image Processing System



Components of an Image Processing System

- 1. Image Sensors** - Two elements are required to acquire digital images. The first is the physical device that is sensitive to the energy radiated by the object we wish to image (*Sensor*). The second, called a *digitizer*, is a device for converting the output of the physical sensing device into digital form.
- 2. Specialized Image Processing Hardware** - Usually consists of the *digitizer*, mentioned before, plus *hardware* that performs other primitive operations, such as an *arithmetic logic unit (ALU)*, which performs *arithmetic and logical operations* in parallel on entire images.
This type of hardware sometimes is called a *front-end subsystem*, and its most distinguishing characteristic is *speed*. In other words, this unit *performs functions that require fast data throughputs* that the typical main computer cannot handle.
- 3. Computer** - The computer in an image processing system is a *general-purpose computer* and can range from a PC to a supercomputer. In dedicated applications, sometimes *specially designed computers* are used to achieve a *required level of performance*.

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Components of an Image Processing System

4. Image Processing Software - Software for image processing consists of *specialized modules* that perform specific tasks. A well-designed package also includes the *capability for the user to write code* that, as a minimum, utilizes the specialized modules.

5. Mass Storage Capability - Mass storage capability is a must in a image processing applications. An image of sized 1024 * 1024 pixels requires one megabyte of storage space if the image is not compressed.

□ **Digital storage for image processing applications** falls into three principal categories:

- ✓ **Short-term storage for use during processing**
- ✓ **Online storage for relatively fast recall**
- ✓ **Archival storage, characterized by infrequent access**

One method of providing *short-term storage* is *computer memory*. Another is by *specialized boards, called frame buffers*, that store one or more images and can be accessed rapidly.

The *on-line storage method*, allows *virtually instantaneous image zoom*, as well as *scroll (vertical shifts) and pan (horizontal shifts)*. On-line storage generally takes the form of *magnetic disks and optical-media storage*. The key factor of on-line storage is *frequent access* to the stored data.

Finally, *archival storage* is characterized by *massive storage requirements* but *infrequent need for access*.

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Components of an Image Processing System

6. Image Displays

The *displays* in use today are mainly *colour (preferably flat screen) TV monitors*. Monitors are driven by the *outputs of the image and graphics display cards* that are an integral part of a computer system.

7. Hardcopy devices

Used for *recording images*, include *laser printers, film cameras, heat-sensitive devices, inkjet units and digital units, such as optical and CD-Rom disks*.

8. Networking

Is almost a default function in any computer system, in use today. Because of the large amount of data inherent in image processing applications the key consideration in *image transmission is bandwidth*.

In dedicated networks, this typically is not a problem, but *communications with remote sites via the internet are not always as efficient*.

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DIGITAL IMAGE REPRESENTATION

- An image can be defined as a two-dimensional function $f(x,y)$
 - x,y : Spatial coordinate
 - $f(x,y)$: the amplitude of any pair of coordinate x,y , which is called the intensity or gray level of the image at that point.
 - x,y and f , are all finite and discrete quantities.
- An image can be defined by a two-dimensional array specifically arranged in rows and columns.
- Digital Image is composed of a finite number of elements, each of which elements have a particular value at a particular location. These elements are referred to as picture elements, image elements, and pixels.
- A Pixel is most widely used to denote the elements of a Digital Image.

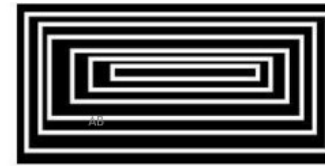
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DIGITAL IMAGE REPRESENTATION

Types of an image

❖ BINARY IMAGE

- The binary image as its name suggests, contain only two pixel elements i.e 0 & 1, where 0 refers to black and 1 refers to white.
- This image is also known as Monochrome.
- Binary images have a format of PBM (Portable bit map)



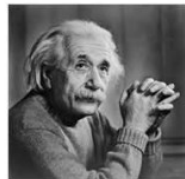
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DIGITAL IMAGE REPRESENTATION

Types of an image

❖ 8 bit COLOR FORMAT(Gray Image)

- It is the most famous image format.
- It has 256 different shades of colors in it and commonly known as Grayscale Image.
- In this format, 0 stands for Black, and 255 stands for white, and 127 stands for gray.
- Format of these images are PGM(Portable Gray Map)
- This format was used initially by early models of the operating systems UNIX and the early color Macintoshes.



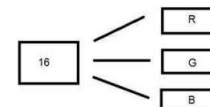
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DIGITAL IMAGE REPRESENTATION

Types of an image

❖ 16 bit COLOR FORMAT

- It is a color image format.
- It has 65,536 different colors in it.
- It is also known as High Color Format.
- In this format the distribution of color is not as same as Grayscale image.
- It is actually divided into three further formats which are Red, Green and Blue (RGB format).
- It is pictorially represented in the image below.



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DIGITAL IMAGE REPRESENTATION

Types of an image

- ❖ How would you distribute 16 into three. If you do it like this,
 - 5 bits for R, 5 bits for G, 5 bits for B
 - Then there is one bit remains in the end.
 - So the distribution of 16 bit has been done like this.
 - 5 bits for R, 6 bits for G, 5 bits for B.
 - The additional bit that was left behind is added into the green bit.
 - Because green is the color which is most soothing to eyes in all of these three colors.
 - This distribution is not followed by all the systems.



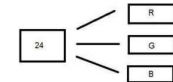
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DIGITAL IMAGE REPRESENTATION

Types of an image

❖ 24 bit COLOR FORMAT

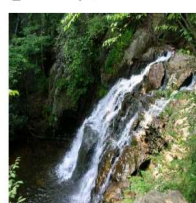
- 24 bit color format also known as true color format.
- In a 24 bit color format, the 24 bits are distributed in three different formats of Red, Green and Blue
- Since 24 is equally divided on 8, so it has been distributed equally between three different color channels.
- Their distribution is like this.
 - 8 bits for R, 8 bits for G, 8 bits for B.
 - Format
 - It is the most common used format. Its format is PPM (Portable pixMap) which is supported by Linux operating system.
 - Windows has its own format for it which is BMP (Bitmap)



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Common image formats

- Common image formats include:
 - 1 sample per point (B&W or Grayscale)
 - 3 samples per point (Red, Green, and Blue)
 - 4 samples per point (Red, Green, Blue, and “Alpha”- Opacity)



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DIGITAL IMAGE REPRESENTATION

Image as a Matrix

- Images are represented in rows and columns, we have the following syntax in which images are represented

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,N-1) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ f(M-1,0) & f(M-1,1) & f(M-1,2) & \dots & f(M-1,N-1) \end{bmatrix}$$

- Every element of this matrix is called image element , picture element , or pixel.
- M and N are positive integers and no restriction is placed on this
- No. of intensity levels L is typically an integer power of 2 : $L=2^k$
- We assume that discrete levels are equally spaced and they are integers in the range $[0,L-1]$.

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DIGITAL IMAGE REPRESENTATION**Image as a Matrix**

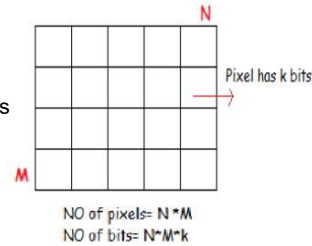
- **M x N pixels** and **G = number of gray levels ($G = 2^k$)**

It is assumed that discrete levels are equally spaced between 0 to G-1 in the gray scale.

- The number of bits(b) required to store a digitized image of size **M x N** is

$$b = M \times N \times k$$

- **k bit image:** An image with 2^k intensity levels
- Eg: 8 bit image: 256 possible discrete intensity values



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DIGITAL IMAGE REPRESENTATION**Image as a Matrix**

- **Dynamic range** of an imaging system is the ratio of maximum measurable intensity to minimum detectable intensity level in the system
- **Contrast:** defined as the difference in intensity between the highest and lowest intensity levels in an image

- An **image processing operation** typically defines a new image in terms of an existing image.

$$g(x, y) = t(f(x, y))$$

- We can transform either the **range of f** or the **domain of f**

$$g(x, y) = f(t_x(x, y), t_y(x, y))$$

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Q. Find the number of bits required to store a 128 x 128 image with 256 gray levels.

Solution:

The number of bits required to store a digitized image of size **M x N** is

$$b = M \times N \times k$$

$$G = 2^k$$

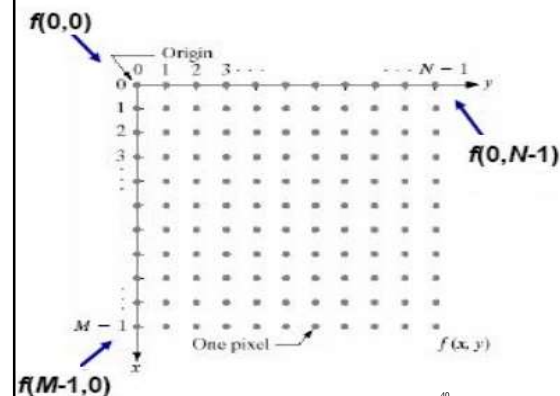
$$256 = 2^8$$

128 x 128 image with 256 gray levels (ie **8 bits/pixel**)

$$128 \times 128 \times 8 = 131072 \text{ bits}$$

$$= 131072 / 8 = 16384 \text{ bytes} \sim 17000 \text{ bytes}$$

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Digital Image Representation**Coordinate Conventions**

L: # of discrete gray levels

$$L = 2^k$$

$$b = M \times N \times k$$

Every pixel has a # of bits.

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Digital Image Representation Coordinate Conventions

- The result of sampling and quantization is a **matrix of real numbers**.
- There are two principle ways to **represent a digital image**: –
- Assume that an **image $f(x,y)$** is sampled so that the resulting image has **M rows and N columns**, then the image is of **size M x N**.
- The values of the coordinates (x,y) are **discrete quantities** (integer values for clarity).
- Usually, the image origin is defined to be at $(x,y) = (0,0)$. The next coordinate values along the first row of the image are $(x,y) = (0,1)$.
- the notation $(0,1)$ is used to signify the **second sample along the first row**.
- these **need not be the actual values of physical coordinates**.
- Note that **x ranges from 0 to M-1**, and **y ranges from 0 to N-1**.

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DIGITAL IMAGE REPRESENTATION Image Sampling and Quantization

- Digitizing the coordinate values is called **sampling**
- Digitizing the amplitude values is called **quantization**
- **Digitization = Sampling + Quantization**
- **Sampling** is done on **X-axis** and **quantization** is done in **Y-axis**.
- ie, **digitizing the gray level resolution of an image** is done in **quantization**.

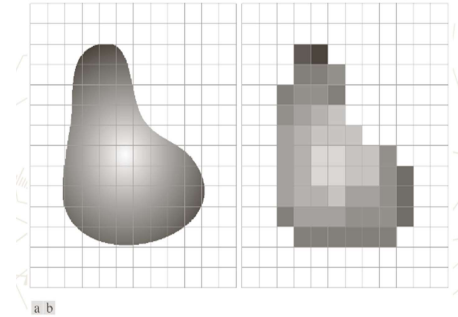
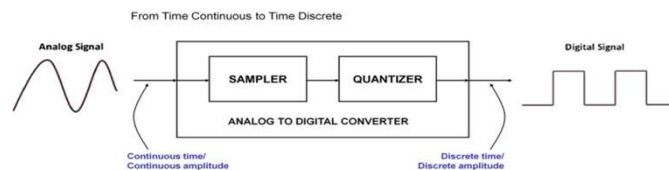


FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

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SAMPLING & QUANTIZATION

To create a digital image, we need to convert the continuous data into digital form. This conversion from analog to digital involves two processes: sampling and quantization.

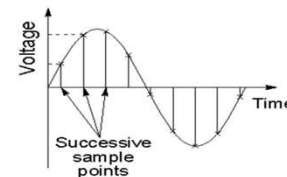


Sampling -> digitization of coordinate values
Quantization -> digitization of amplitude values

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Sampling in Digital Image Processing:

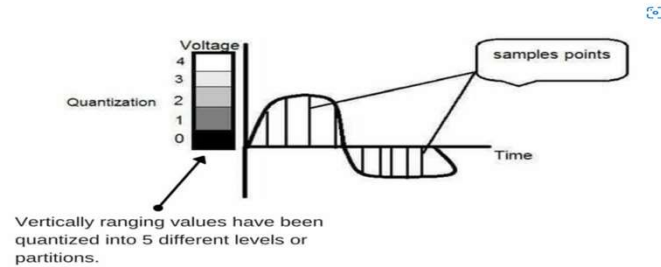
- In this we digitize x-axis in sampling.
- It is done on the independent variable.
- For e.g. if $y = \sin x$, it is done on x variable.



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Quantization in Digital Image Processing:

- It is opposite of sampling as sampling is done on the x-axis, while quantization is done on the y-axis.
- Digitizing the amplitudes is quantization. In this, we divide the signal amplitude into quanta (partitions).

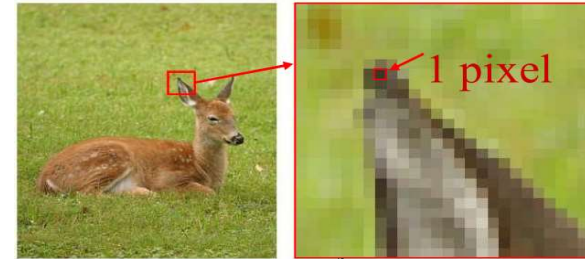


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DIGITAL IMAGE REPRESENTATION

Digitization

- **Digitization** implies that a **digital image** is an approximation of a real scene.
- **Pixel values** typically represent **gray levels, colours, heights, opacities** etc.



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Spatial and Gray Level Resolution

Spatial resolution states that the clarity of an image cannot be determined by pixel resolution (number of pixels in an image).

Spatial resolution can be defined as the **number of independent pixels values per inch**.

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Measuring spatial resolution**Dots per inch**

Dots per inch or DPI is usually used in monitors.

Lines per inch

Lines per inch or LPI is usually used in laser printers.

Pixel per inch

Pixel per inch or PPI is measure for different devices such as tablets , Mobile phones e.t.c.

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Gray level resolution

Gray level resolution refers to the predictable or deterministic change in the shades or levels of gray in an image.

In short , gray-level resolution is equal to the number of bits per pixel (BPP).

The number of different colors in an image depends on the depth of color or **bits per pixel**.

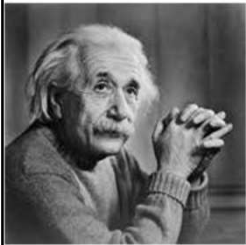
The mathematical relation that can be established between gray level resolution and bits per pixel can be given as.

$$L = 2^k$$

L – number of gray levels
K-bits per pixel

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For example: The image of Einstein is a grayscale image



It is an image with 8 bits per pixel or 8bpp.

Gray-level resolution = $L = 2^k$

Where $k = 8$

$$L = 2^8$$

$$L = 256.$$

It means its gray-level resolution is 256. Or in another way, we can say that this image has **256 different shades of gray**.

The more the bits per pixel of an image, the more is its gray-level resolution.

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Relationship between Pixels**Neighbours of a Pixel**

x-1,y-1	x-1,y	x-1,y+1
x,y-1	x,y	x,y+1
x+1,y-1	x+1,y	x+1,y+1

- A pixel p at coordinates (x,y) has **four horizontal and vertical neighbors** whose coordinates are given by:

(x+1, y), (x-1, y), (x, y+1), (x, y-1)

- This set of pixels, called the **4-neighbors of p**, is denoted by **N4(p)**.
- Each pixel is one unit distance from (x,y) and some of the neighbors of p lie outside the digital image if (x,y) is on the border of the image.

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Relationship between Pixels

Neighbours of a Pixel

$x-1, y-1$	$x-1, y$	$x-1, y+1$
$x, y-1$	x, y	$x, y+1$
$x+1, y-1$	$x+1, y$	$x+1, y+1$

- The four diagonal neighbors of p denoted by $N_D(p)$, have coordinates:
 $(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$
- These points, together with the 4-neighbors, are called the 8-neighbors of p , denoted by $N_8(p)$.
- As before, some of the points in $N_D(p)$ and $N_8(p)$ fall outside the image if (x, y) is on the border of the image.

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Relationship between Pixels

Adjacency and Connectivity

- Let V : a set of intensity values used to define adjacency and connectivity.
 - In a binary image, $V = \{1\}$, if we are referring to adjacency of pixels with value 1.
 - In a gray-scale image, the idea is the same, but V typically contains more elements, for example, $V = \{180, 181, 182, \dots, 200\}$
- If the possible intensity values are 0 – 255, V set can be any subset of these 256 values.

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Relationship between Pixels

Types of Adjacency

4-adjacency:

- Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.

8-adjacency:

- Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

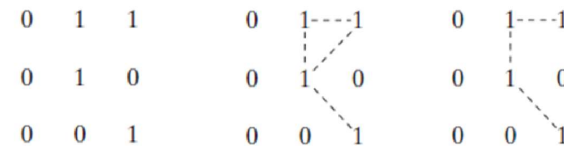
m-adjacency: (mixed)

- Two pixels p and q with values from V are m-adjacent if :
 - q is in $N_4(p)$ or
 - q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V (no intersection)

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Relationship between Pixels

Types of Adjacency



(a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) m-adjacency.

- Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.
- In 8-adjacency way, can find multiple paths between two pixels. While, in m-adjacency, can find only one path between two pixels.
- So, m-adjacency has eliminated the multiple path connection that has been generated by the 8-adjacency.

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Relationship between Pixels A DIGITAL PATH

- A **digital path (or curve)** from pixel p with coordinate (x,y) to pixel q with coordinate (s,t) is a **sequence of distinct pixels with coordinates** $(x_0,y_0), (x_1,y_1), \dots, (x_n,y_n)$
 - where $(x_0,y_0) = (x,y)$ and $(x_n,y_n) = (s,t)$ and
 - pixels (x_i,y_i) and (x_{i-1},y_{i-1}) are adjacent for $1 \leq i \leq n$ is the **length of the path**
 - If $(x_0,y_0) = (x_n,y_n)$, the path is **closed**.
- We can specify 4-, 8- or m-paths depending on the type of adjacency specified.

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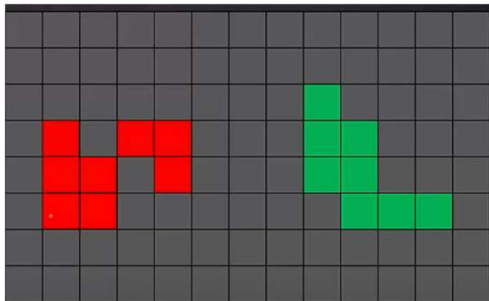
Relationship between Pixels CONNECTIVITY

- Let S represent a subset of pixels in an image, **two pixels p and q are said to be connected in S** if **there exists a path between them consisting entirely of pixels in S** .
- For any pixel p in S , the **set of pixels that are connected to it in S** is called a **connected component of S** .
- If it **only has one connected component**, then set S is called a **connected set**.

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Connected Components of an Image

- Both regions are connected components



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4 adjacency

Binary Image

0	1	0	1
0	0	1	0
0	0	1	0
1	0	0	0

 $v = \{1\}$

Gray Image

54	10	100	8
81	150	2	34
200	201	5	45
7	70	147	56

 $v = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

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8 adjacency

Binary Image

0	1	0	1
0	0	1	0
0	0	1	0
X	0	0	0

 $v=\{1\}$

Gray Image

54	10	100	8
81	150	2	34
200	201	3	45
X	70	147	56

 $v=\{1,2,3,4,5,6,7,8,9,10\}$

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m adjacency

Binary Image

0	1	1
0	1	0
0	0	1

 $v=\{1\}$

Ambiguity

If 4 adjacency and 8 adjacency are possible at the same time priority is for 4 adjacency

0	1	1
0	1	0
0	0	1

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Define 4-adjacency, 8-adjacency and m-adjacency. Consider the image segment shown.

4	2	3	2 q
3	3	1	3
2	3	2	2
2 p	1	2	3

Let $V=\{1,2\}$ and compute the length of the shortest 4-, 8- and m- path between p and q. If a particular path does not exist between these two points, explain why?

4- path between q and p is not possible, because q has no 4 adjacent pixels with values 1 or 2.

8- path between q and p is possible,

4	2	3	2 q
3	3	1	3
2	3	2	2
2 p	1	2	3

Length = 4

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SOLVED EXAMPLE OF PATH LENGTH

Given image -

3	1	2	1 (q)
2	2	0	2
1	2	1	1
10 (p)	0	1	1

Ques Find the shortest 4-, 8-, m- path between p and q.

(i) $V = \{0, 1\}$
(ii) $V = \{1, 2\}$

(i) $V = \{0, 1\}$

4-path:

8-path:

m-path:

(ii) $V = \{1, 2\}$

4-path:

8-path:

m-path:

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Relationship between Pixels REGION AND BOUNDARY

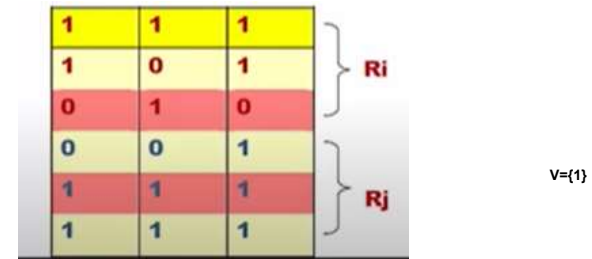
• Region

- Let R be a subset of pixels in an image, we call R a region of the image if R is a connected set.
- Two regions R_i and R_j are said to be adjacent if their union forms a connected set.
- Regions that are not adjacent are said to be disjoint.

• Boundary

- The boundary (also called border or contour) of a region R is the set of points that are adjacent to points in the complement of R .
- Or, border of a region is the set of pixels in the region that have at least one background neighbour.
- Or Set of pixels in the region that have one or more neighbours that are not in the region.

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- R_i and R_j are disjoint w.r.t 4 Adjacency
- 4 path between the two regions does not exist, (so their union is not a connected set)
- R_i and R_j are adjacent w.r.t 8 Adjacency
- Two regions (of 1s) in the figure, are adjacent only if 8-adjacency is used.

Relationship between Pixels REGION AND BOUNDARY

- If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns in the image.
- This extra definition is required because an image has no neighbors beyond its borders
- Normally, when we refer to a region, we are referring to subset of an image, and any pixels in the boundary of the region that happen to coincide with the border of the image are included implicitly as part of the region boundary.

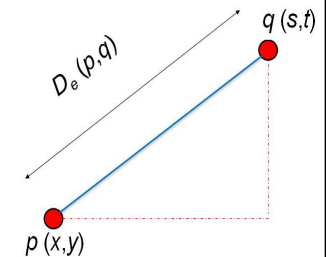
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Relationship between Pixels DISTANCE MEASURES

- For pixels p, q with coordinates $(x, y), (s, t)$ respectively, D is a distance function if:

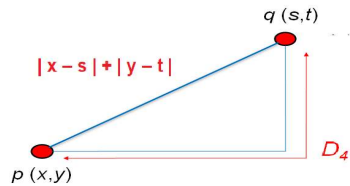
The Euclidean Distance between p and q is defined as:

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$



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Relationship between Pixels DISTANCE MEASURES



- The **D4 distance** (also called **city-block distance**) between p and q is defined as:

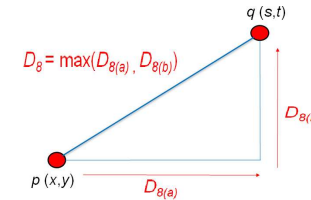
$$(p,q) = |x - s| + |y - t|$$
- The pixels with $D4 = 1$ are the 4-neighbors of (x,y)

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Relationship between Pixels DISTANCE MEASURES

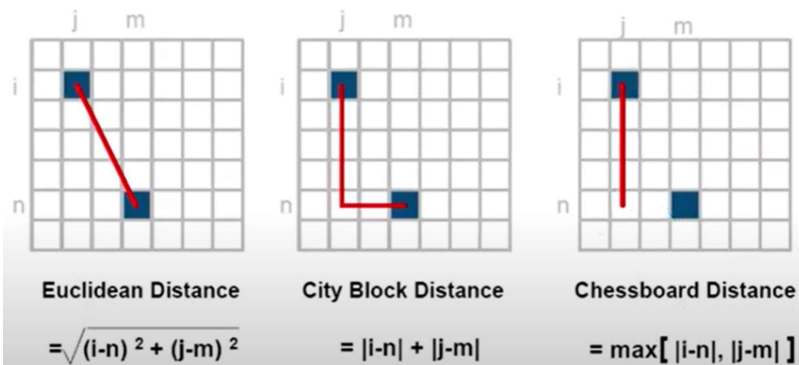
- The **D8 distance** (also called **chessboard distance**) between p and q is defined as:

$$(p,q) = \max(|x - s|, |y - t|)$$



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Relationship between Pixels DISTANCE MEASURES



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Fundamentals of Spatial Domain, Convolution Operation

- **Filtering** creates new pixels with coordinates equal to center of the neighbourhood, and whose value is the result of the filtering operation

1. Linear Spatial filter
2. Non linear Spatial filter

The response(Output) $g(x,y)$, is the sum of the products of the filter coefficients and the image pixel values

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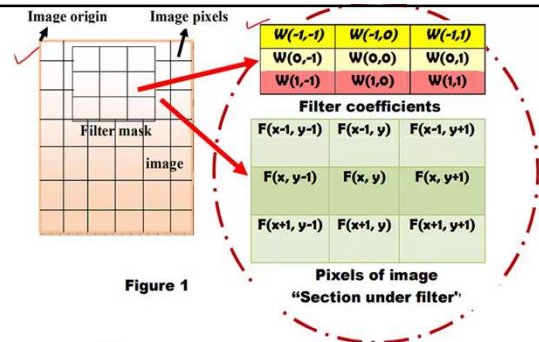


Figure 1

- The center coefficient of the filter $w(0,0)$ aligns with the pixel at location (x,y)

$$g(x,y) = w(-1,-1)f(x-1,y-1) + w(-1,0)f(x-1,y) + \dots + w(0,0)f(x,y) + \dots + w(1,1)f(x+1,y+1)$$

- General mask of size $m \times n$
- Assume $m = 2a + 1$ $n = 2b + 1$

In general, linear spatial filtering of an image of size $M \times N$ with a filter of size $m \times n$ is given by the expression:

$$g(x,y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s,t) \cdot f(x+s, y+t)$$

Where x and y are varied so that each pixel in w visits every pixel in f .

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Spatial Correlation and convolution

- Correlation** : The process of moving a filter mask over the image and computing the sum of products at each location.
- Convolution** : The process of rotating a filter mask by 180° and moving the filter mask over the image and computing the sum of products at each location.

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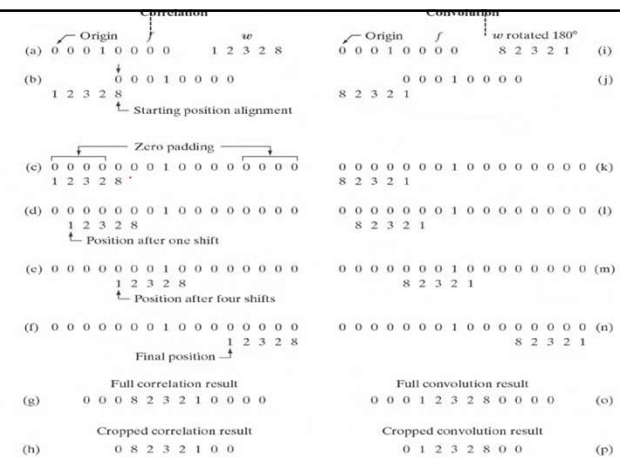


FIGURE 3.29 Illustration of 1-D correlation and convolution of a filter with a discrete unit impulse. Note that correlation and convolution are functions of displacement.

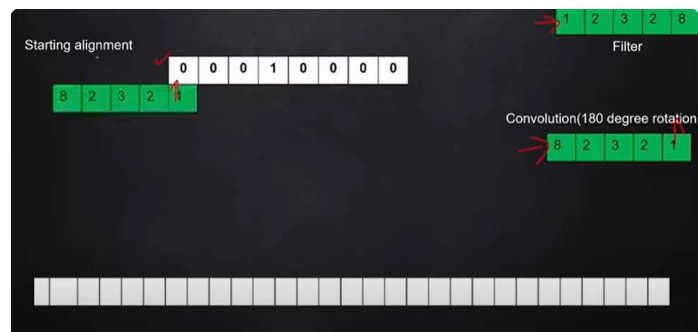
Notes:

- There are parts of the functions (images) that do not overlap (the solution of this problem is pad f with enough 0s on either side to allow each pixel in w to visit every pixel in f).
- If the filter is of size m, we need (m-1) 0s on either side of f.
- The first value of correlation is the sum of products of f and w for the initial position (Figure 2.c). (The sum of product =0) this corresponds to a displacement $x=0$
- To obtain the second value of correlation, we shift w are pixel location to the right (displacement $x=1$) and compute the sum of products (result =0).
- The first nonzero is when $x=3$, in this case the 8 in w overlaps the 1 in f and the result of correlation is 8.
- The full correlation result (figure 2.g) - 12 values of x
- To work with correlation arrays that are the same size as f, in this case, we can crop the full correlation to the size of the original function. (Figure 2.h).

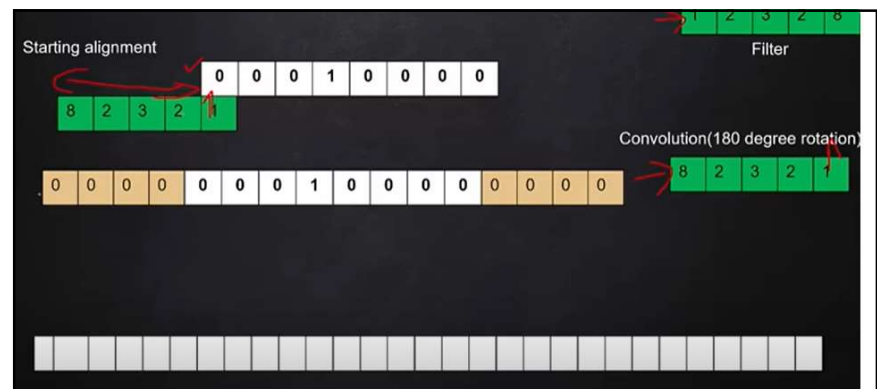
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- The result of correlation is a copy of w, but rotated by 180 degree.
- The correlation with a function with a discrete unit impulse yields a rotated version of the function at the location of the impulse.
- The convolution with a function with a discrete unit impulse yields a copy of that function at the location of the impulse

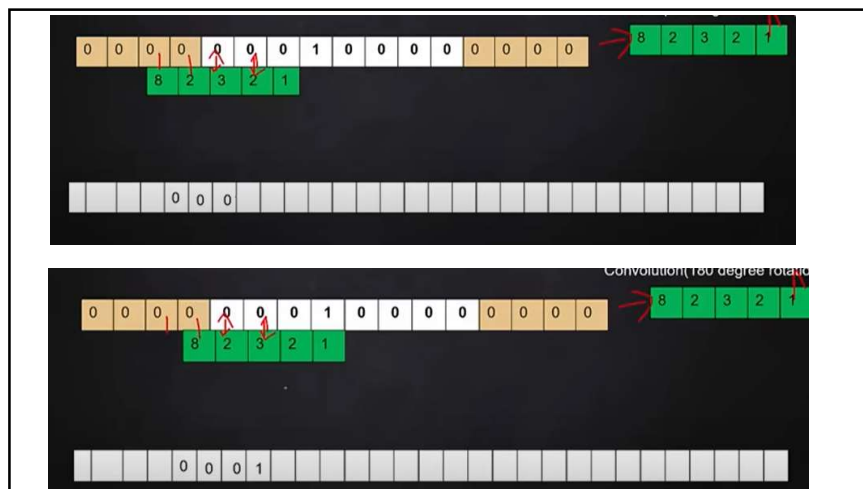
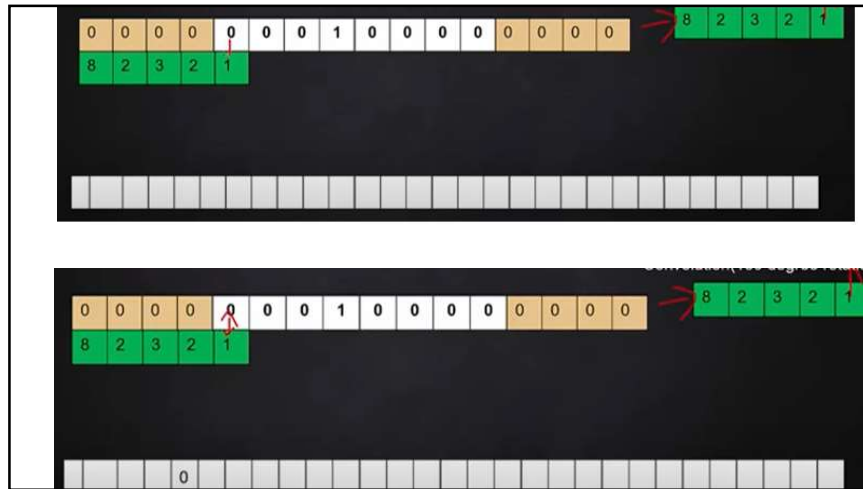
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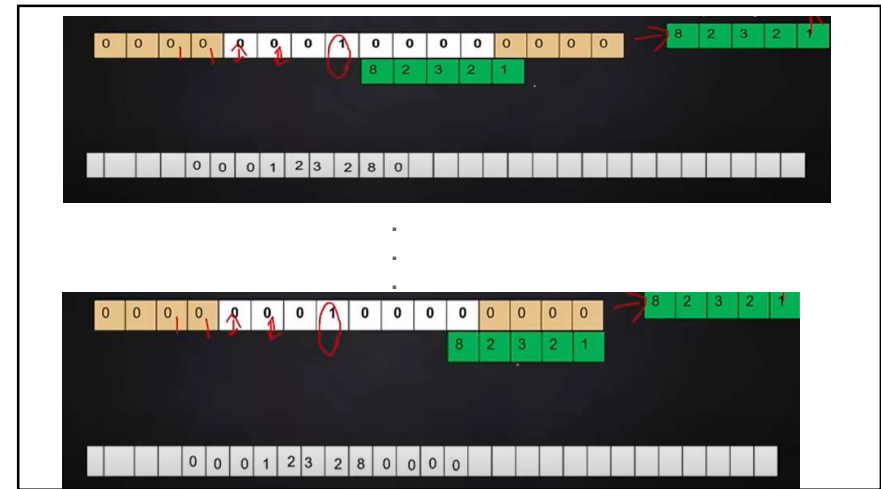


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Result after removal of padded zeros ,



Correlation and Convolution with Images

- With a image size of $m \times n$, we pad the image with a minimum $m-1$ rows of 0s at the top and the bottom, and $n-1$ columns of 0s on the left and right

- Correlation of a filter $\underline{w}(x,y)$ of size $m * n$ with an image $f(x,y)$ denoted as

$$\underline{W}(x,y) \circ f(x,y) = \sum_a \sum_b w(s,t) f(x+s, y+t)$$

$s=-a \quad t=-b$

- In similar manner, the convolution of $\underline{w}(x,y)$ and $f(x,y)$ denoted by $\underline{w}(x,y) * f(x,y)$ is given by:

$$\underline{W}(x,y) * f(x,y) = \sum_a \sum_b w(s,t) f(x-s, y-t)$$

$s=-a \quad t=-b$

Where the minus sign on the right flip (rotate by 180°)

(We can flip and shift either f or w)

